FINAL YEAR PROJECT · PHASE 1 · ZEROTH REVIEW

EYEQ: A Hybrid Quantum–Classical Eye-Gaze Typing System on Commodity Webcams

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PROJECT GUIDE

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Introduction

- Text entry still assumes working hands and a physical keyboard — excluding users with motor impairments, and failing in hands-occupied or sterile settings such as operating theatres and laboratories.
- Research-grade eye tracking already solves the sensing problem, but it is priced for laboratories. Wearable trackers such as Tobii Pro Glasses are sold by quotation, with a published entry price near ₹ 9.6 lakh for the previous generation.
- Even a consumer screen-mounted tracker such as the Tobii Eye Tracker 5 costs about ₹ 48,000 — still an unrealistic purchase for someone who needs it purely in order to communicate.
- EYEQ runs entirely on a commodity webcam of roughly ₹ 1,900 — hardware most users already own — with no infrared hardware, no client build step, and all gaze computation kept on the user's own device.
- Large, gaze-friendly key clusters make targets easy to hit despite noisy gaze, but several letters share one region, so distinct words collapse onto the same cluster code. A quantum conditional selection rule resolves that ambiguity, improving the accuracy with which the intended word is committed.



Objectives

1

Gaze estimation on ordinary hardware

To estimate gaze from MediaPipe iris and head-pose landmarks through a per-user ridge-regression calibration, using a commodity webcam with no infrared hardware.

2

Reliable, fatigue-aware selection

To commit characters by dwell selection over large, gaze-friendly key clusters that remain easy to hit under realistic gaze noise.

3

Higher accuracy on the intended word

To apply a quantum conditional selection rule that fuses gaze evidence with linguistic context, improving the accuracy with which the intended word is committed.

4

A guarantee, not a hope

To ensure through a consistency-preserving oracle that every decoded output re-encodes to exactly the clusters gazed at — never an off-target word — while staying within the webcam frame budget.

Abstract

EYEQ is an eye-gaze typing system that runs entirely on a commodity webcam — no infrared hardware and no client build step. Gaze is estimated by mapping **MediaPipe iris and head-pose landmarks** through a **per-user ridge-regression calibration**, and characters are committed by **dwell selection** over large, gaze-friendly key clusters. Clustering makes targets easy to hit but renders the input ambiguous: several letters share one region, so distinct words collapse onto the same cluster code. To resolve this ambiguity accurately and in real time, EYEQ applies a **quantum conditional selection rule** that fuses the gaze evidence with linguistic context and commits the most probable candidate word. A **consistency-preserving oracle** guarantees that every decoded output re-encodes to exactly the gazed clusters, never an off-target word. The rule is realised as a lightweight classical state-vector simulation, keeping the full pipeline within the webcam frame budget.

Literature Survey

Title	Author	Methodology	Addresses	Limitation
Augmentative and Alternative Communication Using Eye Tracking and Word Recommendation Using Language Models (2025)	B. Waideman and P. T. Aquino Jr.	Webcam CNN regressing screen (x, y) from face, both eyes and head pose; nine-point calibration; trigram language model for word recommendation	Low-cost AAC keyboard; 31.58 px gaze RMSE; 7.3 s per character; 93% precision	Language model only predicts the next word — it cannot correct a mis-selected key; training needed dual V100 GPUs
Dwell Time Adaptive Three-Way Text Entry by Gaze (2019)	D. Li, L. Zhang, W. Hu, H. Li and X. Zhou	Three-way decision splitting each key into positive, boundary and negative regions; adaptive dwell time driven by typing speed	Midas-touch problem and ambiguity when a fixation lands between two keys	Requires a commercial SMI eye tracker; boundary cases push the choice back to the user instead of resolving it
Empowering Quadriplegic Patients: A Computer Vision Based Approach for Keyboard Control Using Eye Gaze and Blink (2023)	S. T. S. Rafid, A. J. Ajwad and A. K. M. A. Azad	MediaPipe Face Mesh with 478 landmarks; iris centre via minEnclosingCircle; horizontal and vertical ratios; five-point calibration; 2×4 grid	Webcam-only typing for quadriplegic users; 14.53 characters per minute	Coarse 2×4 grid classification rather than continuous gaze regression; no linguistic error correction
Computer Vision Based Eye Gaze Controlled Virtual Keyboard for People with Quadriplegia (2021)	M. R. Islam, M. S. Rahman and A. Sarkar	HOG and SVM face detection; dlib landmarks; sclera-area gaze ratio; 40-key scanning keyboard selected by blink	Assistive keyboard at $20 hardware cost; 9.50 characters per minute	Sequential scanning is inherently slow; fixed gaze-ratio thresholds with no per-user calibration
Eye-Gaze Controlled Virtual Keyboard Using Blink Detection for Accessible Hands-Free Communication (2025)	B. G. Varshini, A. Manimegalai, S. D. Abirami and S. Kalaiselvi	dlib 68-point landmarks; Eye Aspect Ratio blink detection at a 0.25 threshold; cycling highlight over letter groups	Hands-free entry at 17 WPM with 92.5% detection accuracy	Selection is by scanning rather than true gaze pointing; 120 false-positive blinks across 5,000 frames

References

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Thank You